

PhonePe Transaction Analysis

Power BI Project Report

End-to-end data cleaning, modeling, DAX measures and interactive dashboard development

1. Project Overview

PhonePe Transaction Analysis is an end-to-end Power BI project focused on understanding transaction volume, transaction value, users, payment status, services, age segments and time-based patterns. The complete workflow—from data cleaning through dashboard development—was performed in Power BI.

The supplied cleaned Excel workbook contains the prepared transaction, user and date tables, while the supplied PBIX contains the report model, DAX measures, KPI cards, slicers, analytical visuals and tooltip pages.

2. Project Objectives

- Clean and prepare PhonePe transaction and user data for analysis.
- Replace inconsistent values, remove duplicates and blank rows, create conditional columns and assign suitable data types.
- Create a dedicated DAX date table for time-based analysis.
- Create a dedicated measure table to organize reusable DAX calculations.
- Analyze transaction count, transaction amount and payment success performance.
- Analyze service-wise, age-segment-wise and weekday/weekend patterns.
- Build an interactive Power BI dashboard using KPIs, charts and slicers.
- Use tooltip pages to provide additional information without overcrowding the main dashboard.

3. Dataset Structure

Table	Rows	Columns	Purpose
All_Transactions	300,000	8	Core transaction-level data
All_Users	107,658	5	User/customer attributes
Date_Table	365	9	Calendar and time-analysis fields
measures (2)	—	—	Dedicated DAX measure table in the PBIX

All_Transactions fields: Transaction_ID, Amount, User_ID, Service, Service Type, Payment_Status, Reason and Date.

All_Users fields: User_ID, Name, Age, Join_Date and age segment.

Date_Table fields: Date, year, month no., month, quarter, week, day no., weekend and month (groups).

4. Data Cleaning and Preparation

The project includes the following preparation activities:

- Replaced values to standardize or correct categorical data.
- Removed duplicate records.
- Added a conditional column for derived classification.
- Removed blank rows.
- Changed column data types to appropriate analytical types.
- Created a DAX date table for calendar-based filtering and analysis.
- Created a dedicated measure table for organizing DAX calculations.

5. Data Model

The PBIX model contains All_Transactions, All_Users, Date_Table and a dedicated measures table. The report configuration confirms that transaction fields are used for transaction/service analysis, user fields are used for age-segment analysis, and Date_Table fields are used for month and weekend analysis.

The dedicated date table provides a structured calendar for time-based reporting, while the separate measure table keeps DAX calculations organized and easier to maintain.

6. DAX Measures and KPIs

The PBIX uses a dedicated measure table named “measures (2)”. The following measures are directly referenced by the report:

Measure	Purpose in dashboard
total no. of trans	Transaction-count KPI, trend analysis
total transaction amount	Transaction-value KPI and amount-based charts
total users	User KPI
success rate	Payment-success KPI
total trans mom%	Month-over-month transaction KPI
trans amount mom%	Month-over-month transaction amount KPI

Dataset-level validation : 300,000 transactions, ₹3,474,321,934.16 total transaction amount, 100,761 distinct transacting users, and 96.00% successful transactions.

7. Dashboard Design and Visuals

The PBIX contains one main report page named “Page 3” and two tooltip pages named “tooltip1” and “tooltip2”. The main page is configured at 1920 × 1080.

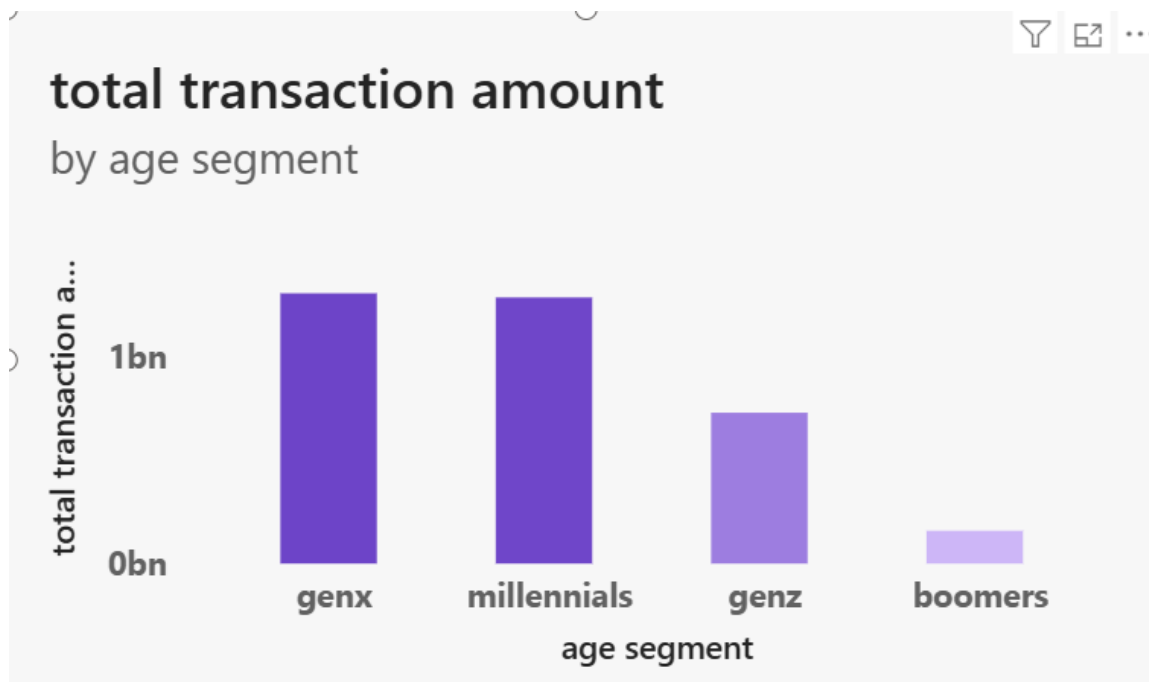
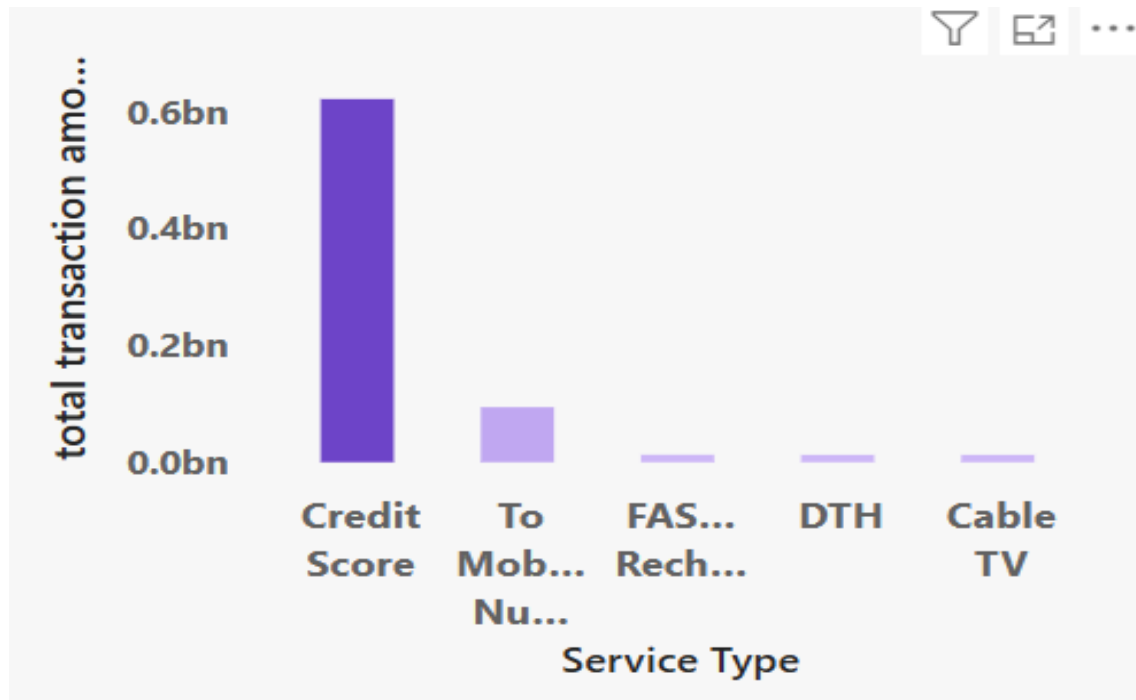
Element	Implemented design
Branding	PhonePe heading, logo/image assets and purple branded typography
KPI cards	Total transaction, total transaction amount, total users and success rate
Trend	Area chart using month with transaction count and transaction amount
User segmentation	Donut chart for users by age segment
Service analysis	Column chart for transaction amount by service
User-level analysis	Column chart using user names and transaction amount
Weekend analysis	Donut chart for transaction amount by weekday/weekend
Slicers	Payment Status and Month
MoM indicators	Transactions MoM% and transaction amount MoM% cards
Tooltip 1	Service Type vs total transaction amount
Tooltip 2	Age segment vs total transaction amount



8. Tooltip Pages

Two dedicated tooltip pages are present in the PBIX. They add contextual detail when the user hovers over related visuals and help keep the main dashboard clean.

- tooltip1 contains a column chart showing total transaction amount by Service Type.
- tooltip2 contains a column chart showing total transaction amount by age segment.



9. Key Findings

- The transaction data contains 300,000 records covering 1 January 2024 through 30 December 2024.
- Total transaction value is approximately ₹3.47 billion.
- Money Transfer has the highest transaction volume with 150,000 records.
- Loans contributes the highest transaction amount at approximately ₹2.53 billion.
- There are 9,980 failed and 2,027 pending transaction records.
- The user table contains 107,658 user records, while 100,761 distinct users appear in the transaction table.
- Gen X is the highest transaction-value age segment in the supplied data.

10. End-to-End Project Workflow

1. Import the source PhonePe data into Power BI.
2. Clean and transform the data using Power Query.
3. Replace values and remove duplicates and blank rows.
4. Create conditional/derived columns and correct data types.
5. Create the DAX date table.
6. Create and organize measures in the dedicated measure table.
7. Build KPI cards and analytical visuals.
8. Add slicers for interactive filtering.
9. Create tooltip pages for additional context.
10. Apply PhonePe-oriented formatting and dashboard design.
11. Validate the report against the cleaned dataset.

11. Project Strengths

- Covers the complete Power BI workflow instead of only visualization.
- Uses a dedicated calendar table for time-based analysis.
- Uses a dedicated measure table for clean DAX organization.
- Combines KPIs, trend analysis, categorical analysis, segmentation and slicers.
- Uses tooltip pages to provide additional information without adding visual clutter.
- Uses consistent PhonePe-inspired branding.
- The cleaned Excel workbook is reusable outside the PBIX.

12. Tools and Technologies

Tool	Role
Microsoft Power BI	Data transformation, modeling, DAX and dashboard development
Power Query	Cleaning, replacement, duplicate/blank removal and type transformation
DAX	Date table and analytical measures
Microsoft Excel	Prepared cleaned dataset storage and validation

13. Conclusion

PhonePe Transaction Analysis demonstrates a practical business-intelligence workflow in Power BI. The project covers data cleaning, structured modeling, DAX calculations, KPI development, interactive visualization and tooltip-based exploration. It can be presented as a portfolio case study showing how transaction data is transformed into business insights through Power Query, data modeling, DAX and dashboard design.